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ENGINEERING DIPLOMA THESIS

Title of the thesis: Design and construction of a small anthropomorphic robot

Title of the thesis (in Polish): Projekt i budowa małego robota antropomorficznego

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ABSTRACT

This thesis details the design, construction, and validation of a programmable, small-scale anthropomorphic manipulator. The system is engineered for a 500 mm reach and a 1 kg maximum payload, featuring a 5-axis kinematic structure with an automatically calibrating, interchangeable end-effector. The mechanical assembly utilizes a hybrid construction of custom 3D-printed components with reinforced brass inserts, precision-cut aluminum sheets, and V-slot profiles.

To ensure precision and safety, the manipulator employs five NEMA17 stepper motors integrated into a closed-loop control system via rotary encoders. A key technical contribution is the implementation of active collision detection with adjustable force thresholds, facilitating safe interaction between the robot and its environment.

The control architecture is powered by an ESP32 dual-core microcontroller running custom firmware, which interfaces with a dedicated Python API over Wi-Fi or Ethernet. Supporting hardware includes a custom PCB for power distribution and sensor integration. The final prototype demonstrates a modular approach to robotics, providing a scalable platform for precise manipulation and future iterative upgrades.

Keywords:

Field of science and technology in accordance with OECD requirements:

LIST OF IMPORTANT SYMBOLS AND ABBREVIATIONS

Basic Physical Quantities and Mathematical Expressions

j	$[\cdot]$	–	imaginary unit
∇	$[\cdot]$	–	nabla operator
$\vec{i}_x, \vec{i}_z, \vec{i}_y$	$[\cdot]$	–	unit vectors of the Cartesian coordinate system
$\vec{F}$	[N]	–	force
t	[s]	–	time
m	[kg]	–	mass
R	$[\Omega]$	–	resistance
S	$[\cdot]$	–	scattering matrix

Physical and mathematical constants

$\pi \approx 3.14159$	$[\cdot]$	–	number π
$g \approx 9.80665$	$[\frac{\mathrm{m}}{\mathrm{s}^2}]$	–	gravitational acceleration

Terminology used in the thesis

`\section{}`	–	creates a chapter title
`\subsection{}`	–	creates a subsection title
`\cite{}`	–	reference to a bibliography item
`\ref{}`	–	reference to a figure, table, or equation
`\eqref{}`	–	reference to an equation (with round bracket)
`\begin{figure}` ... `end{figure}`	–	inserting a graphic object
`\begin{table}` ... `end{table}`	–	inserting a table
`\begin{equation}` ... `end{equation}`	–	inserting a numbered equation
`\textbf{}`	–	boldface text
`\emph{}`	–	italic emphasis
`\footnote{}`	–	insert a footnote
`\includegraphics{}`	–	insert an image
`\label{}`	–	object label
`\item`	–	list item

TABLE OF CONTENTS

Abstract 2

List of Important Symbols and Abbreviations 3

Table of contents 4

Bibliography 4

List of figures 5

List of tables 6

LIST OF FIGURES

LIST OF TABLES